N.S. LABORATORY SDN. BHD.

NS-OTC, 500mg/g,

Powder for use in drinking water

Active Ingredient:
Each gram contains
Oxytetracycline (as hydrochloride)---500mg

Product Description:
Yellow powder which can be dissolved in animal drinking water.

Dosage form:
Powder for use in drinking water

Pharmacodynamics:
The oxytetracycline links reversibly to the ribosomal subunit 30S receptors, this leading to a blockage of the union between aminoacyl-tRNA to the site corresponding to the mRNA-ribosome complex messenger. It results in an inhibition of the protein synthesis and inhibits bacterial growth. The mainly bacteriostatic activity of oxytetracycline involves uptake of the substance into the bacterial cell which occurs by both passive and active diffusions. The main mechanism of resistance is due to the possible presence of a R factor responsible for a decrease in the active transport of oxytetracycline. Oxytetracycline is a broad-spectrum antibiotic. It is mainly active against Gram positive and Gram negative bacteria, aerobic and anaerobic, as well as against Mycoplasma, the Chlamydia and Rickettsiae. Acquired resistance to oxytetracycline has been reported. This resistance is usually of plasmid origin. Cross-resistance to other tetracyclines is possible. The prolonged or repeated use of oxytetracycline as well as continuous treatment with low doses of oxytetracycline may also cause increased resistance to other antibiotics due to potential co-resistance with other antimicrobials. Four resistance mechanisms acquired by microorganisms against tetracyclines in general have been reported: decreased accumulation of tetracyclines (decreased permeability of the bacterial cell wall and active efflux), protein protection of the bacterial ribosome, enzymatic inactivation of the antibiotic and rRNA mutations (preventing the tetracycline binding ribosome).

Pharmacokinetics:
The oral absorption of oxytetracycline is low. The mean values of oral absorption of oxytetraycline are 3-5% in pigs and 48% in turkeys. This bioavailability can be reduced in the presence of food in the stomach as oxytetracycline leads to the formation of insoluble chelates with divalent or trivalent cations (Mg, Fe, Al, Ca). In pigs, the influence of food is negligible on the bioavailability of oxytetracycline which is less than 5%. The oxytetracycline binds variably to plasma proteins according to the species (75%). Its distribution is large. The oxytetracycline diffuses throughout the body, the highest concentrations have been found in the kidneys, liver, spleen and lungs. The oxytetracycline crosses the placental barrier. Oxytetracycline is excreted unchanged mainly via urine. It is also excreted via bile but a high proportion of oxytetracycline is reabsorbed by the small intestine (enterohepatic cycle).

Indication:
In chickens (broilers, breeding hens) and pigs- Treatment and metaphylaxis at the group level of septicaemia, respiratory and gastrointestinal infections caused by bacteria sensitive to oxytetracycline. The presence of disease in the herd/group should be established before the product is used.

Mode of Administration:
Oral administration via drinking water.

Recommended Dosage:
The uptake of medicated drinking water depends on the clinical and physiological conditions of the animals. In order to obtain the correct dosage, the concentration of oxytetracycline must be adjusted by calculating the required mean daily water consumption. The duration of treatment is 3 to 5 days, for both chickens and pigs. Dosing is presented in the following table:

Species	Expressed in mg of oxytetracycline / kg of bodyweight/ day	Expressed in mg of ORAL POWDER/ 10 kg of bodyweight/ day	Estimated water consumption (L/ kg of bodyweight)	Expressed in mg of ORAL POWDER/ L of drinking water
Pigs	20 mg	400 mg of ORAL POWDER	1L/ 10 kg of bodyweight	400 mg of ORAL POWDER
Chickens	20 mg	400 mg of ORAL POWDER	1L/ 5 kg of bodyweight	200 mg of ORAL POWDER

Based on the recommended dose, and the number and weight of the animals to be treated, the exact daily amount of oxytetracycline should be calculated according to the following formula:

$$\frac{x\ mg\ oxytetracycline/kgbodyweight/day\ x\ Mean\ body\ weight\ (kg) of\ the\ animals\ to\ be\ treated}{Mean\ daily\ water\ consumption\ (l)\ per\ animal} = x\ mg\ oxytetracycline/\ l\ of\ drinking\ water$$

To ensure a correct dosage, body weight should be determined as accurately as possible to avoid under dosing. The use of suitability calibrated weighing equipment is recommended if part packs are used. The daily amount is to be added to the drinking water such that all medication will be consumed in 24 hours. Medicated drinking water should be freshly prepared every 24 hours.
For full advantages of solubility qualities, it is recommended to prepare a concentrated pre-solution – approximately 400 grams product per litre drinking water – and to dilute this further to therapeutic concentrations if required. Alternatively, the concentrated solution can be used in a proportional water medicator.

Contraindications:
Do not use in cases of hypersensitivity to oxytetracycline or any other substance from tetracyclines group. Do not use in cases of known tetracycline resistance.

Special Warnings for each Target Species:
None.

Special Precautions for Use in Animals:
This powder should be dissolved in water, before use. Use of the product should be based on susceptibility testing of bacteria isolates from the animal. If not possible, therapy should be based on local (regional, farm level) epidemiological information about susceptibility of the target bacteria. Official, national and regional antimicrobial policies should be taken into account when the product is used. Use of the product deviating from the instructions given may increase the prevalence of bacteria resistant to the oxytetracycline and may decrease the effectiveness of treatment with tetracyclines, due to the potential for cross-resistance. Prolonged or repeated use should be avoided as these practises can enforce development and spread of the bacterial resistance. This is particularly likely in Enterobacteria and *Salmonella spp.*, many of which are already resistant. As eradication of the target pathogens may not be achieved, medication should be combined with good management practices, e.g. good hygiene, proper ventilation, no overstocking. Extensive resistance to oxytetracycline has been recognised in porcine and poultry isolates of strains form *E. coli, Salmonella spp., Campylobacter spp., and Enterococcus spp*. The product should only be used where culture and sensitivity testing have demonstrated that it is likely to be effective. Sick animals may have a reduced appetite and an altered drinking pattern and should, if necessary, be medicated parenterally.

Special Precautions to be taken by the Person Administering the Veterinary Medicinal Product to Animals:
People with known hypersensitivity to tetracyclines should avoid contact with the veterinary medicinal product. Avoid inhaling dust when handling the product until complete solubilisation in water. Use in a well-ventilated area away from draughts. Avoid contact with skin and eyes. Personal protective equipment consisting of latex and nitrile gloves, eye protection dust mask and suitable protective clothing should be worn when handling the veterinary medicinal product. In case of accidental eye or skin contact, rinse the affected area with large amounts of clean water. If irritation occurs, seek medical advice immediately and show the label to the physician. Swelling of the face, lips or eyes, or difficulty with breathing are more serious symptoms and require urgent medical attention. Wash hands and contaminated skin immediately after handling the product. Do not smoke, eat or drink while handling the product.

Interactions with other Medicaments:
Divalent or trivalent cations (Mg, Fe, Al, Ca) may chelate with tetracyclines. The tetracylines should not be administered with antacids, gels containing aluminium, preparations containing vitamins or minerals as insoluble complexes will be formed, which decreases the absorption of the antibiotic.

Usage during Pregnancy and Lactation:
Laboratory studies in animals have not produced any evidence of embryotoxicity or teratogenic effects. In mammals, oxytetracycline passes the placental barrier, resulting in staining of teeth and slow foetal growth. Tetracyclines are found in breast milk. Use only according to the benefit/risk assessment by the responsible veterinarian.

Adverse Effects:
As for all other tetracyclines, side effects such as gastro-intestinal disorder and less frequently, allergic and photosensitivity reactions are very rare according to pharmacovigilance data.

Overdose and Treatment:
None known.

Withdrawal Periods (Meat and Offal):
Chickens- 7 days
Pigs- 7 days
Eggs- do not use in laying birds producing eggs intended for human consumption.

Storage Condition:
Store in cool, dry place (< 30ºC).

Shelf Life:
24 months

Shelf Life after Opening:
Discard any unused powder immediately after opening

Shelf Life after Dilution/Reconstitution:
24 hours

Packaging Available:
100g/packet, 100packets/carton, 10kg/carton;
500g/packet, 30packets/carton, 15kg/carton

Manufacturer and Registration Holder:
N.S. Laboratory Sdn. Bhd.
54, Jalan S2C2, Green Technology Park, Seremban 2, 70300 Seremban, Negeri Sembilan, Malaysia.

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18/12/2022